Krishnaa Vadivel | Semiconductor Physics | Integrated Photonics | Thin-Film Fabrication

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Summary

Electrical engineering PhD researcher with Cornell and Yale ECE training across thin-film fabrication, semiconductor physics, integrated photonics, device simulation, and scientific computing. Experience spans cleanroom GaN device development, optical and materials characterization, light-matter modeling, and distributed HPC software for semiconductor and photonic systems.

Experience

Yale University, Electrical Engineering
PhD Research Assistant *2021–present*

- Developed GaN-based power-electronics devices in cleanroom and MOCVD environments during the early PhD, including selective-area etching techniques and materials characterization using SEM, cathodoluminescence, and Raman spectroscopy.
- Modeled VCSEL emission and related light-matter interactions involving coupled photon, phonon, and electron dynamics for semiconductor and photonics problems.
- Built distributed HPC software for exciton-polaron and light-matter interaction modeling using first-principles methods, quantum many-body theory, and accelerator-enabled scientific computing.
- Used agentic ML tooling to generate first-principles software inputs from journal literature and technical documentation.
- Co-authored a 2025 *Journal of the American Chemical Society* paper and delivered APS presentations in 2024, 2025, and 2026 on self-trapped exciton formation and related excited-state materials physics.

Probe Technologies Inc. / Weatherford Wireline Services
Software and Embedded Systems Engineer *2019–2021*

- Parallelized CUDA software for analyzing and visualizing acoustic and nuclear well-tool data used in engineering diagnostics and field interpretation.
- Applied dolfinx-based finite-element analysis to frequency studies of steel pipes in well environments.
- Supported instrumentation-oriented engineering around deployed well tools, connecting field data, sensor systems, and numerical analysis.

FindLight
Photonics Technical Writer *2018–2019*

- Wrote technical content on lasers, integrated optics, and optoelectronic components for engineering-facing audiences.

Selected Projects

Integrated Photonics Band-Pass Filter: Designed a Meep-based integrated photonics band-pass filter, then fabricated and tested the device with an erbium-doped fiber laser setup.

Semiconductor Simulation: Developed numerical modeling of semiconductor electrostatics, drift-diffusion transport, and pn-junction-style device behavior.

VCSEL Light-Matter Modeling: Built simulation tools for VCSEL emission and coupled photon, phonon, and electron interactions in semiconductor systems.

Equivariant Scientific ML: Developed PyTorch Geometric equivariant neural networks for material energy prediction and optical-property prediction.

Education

Yale University
PhD, Electrical Engineering *2021–present*

Yale University
MPhil, Electrical Engineering *2023*

Yale University
MS, Electrical Engineering *2023*

Cornell University
MEng, Electrical and Computer Engineering *2017–2018*

Cornell University
BS, Electrical and Computer Engineering *2013–2017*

Technical Skills

Fabrication / Characterization: Cleanroom processing, MOCVD, SEM, cathodoluminescence, Raman spectroscopy, laser-bench optical testing

Devices / Photonics: Semiconductor physics, thin films, integrated photonics, lasers, VCSEL modeling, light-matter interaction

Simulation: Semiconductor device simulation, FDTD, finite-element analysis, first-principles and quantum many-body modeling

Machine Learning: PyTorch, PyTorch Geometric, equivariant graph neural networks, optical- and materials-property prediction

Software / HPC: Python, C, C++, CUDA, Linux, MPI, OpenMP, PETSc, SLEPc, Docker